

SHELLINGS OF SPHERES AND POLYTOPES

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Introduction. Bruggesser and Mani [7] proved the boundary complex $B(P)$ of a polytope P always admits a shelling. By constructing appropriate linear functionals on certain polytopes dual to P , shellings are here produced that satisfy strong conditions on the order of appearance of P 's facets. Also included are some results on shellings of more general complexes.

Our terminology is adapted from Bruggesser and Mani [7] and Grünbaum [9]. A *polytope* is the convex hull of a finite set of points and a *complex* is a finite set $\mathcal{C}$ of polytopes (in a finite-dimensional Euclidean space) such that $\mathcal{C}$ includes all faces of its members and the intersection of any two members is a face of each. The members are $\mathcal{C}$'s *cells* and the maximal members are its *facets*. Prefixes are used to indicate dimension. The $(d - 1)$ -faces of a d -polytope P (the maximal members of $\mathcal{B}(P)$) are P 's *facets*, and we rely on context to distinguish between the two different uses of the term *facet*. A complex is *simplicial* [resp. an *n-complex*] provided that its facets are all simplices [resp. n -cells]. The underlying set of a complex $\mathcal{C}$ is denoted by $\cup \mathcal{C}$, the notation $|\mathcal{C}|$ being used for cardinality.

A *semishelling* of an n -complex $\mathcal{C}$ is an ordering $(F_1, \dots, F_m)$ of $\mathcal{C}$'s facets such that for $1 < j \leq m$ the intersection $F_j \cap (F_1 \cup F_2 \cup \dots \cup F_{j-1})$ is either a topological $(n - 1)$ -ball or the entire relative boundary of F_j ; when the first alternative holds for all $j < m$ the semishelling is a *shelling*. Bruggesser and Mani [7] proved the boundary complex of a polytope always admits a shelling and any two facets can be specified as F_1 and F_m . McMullen [13] and Klee [12] used additional restrictions on the shelling order to establish other properties of polytopes; in particular, shelling was essential in McMullen's proof of the upper bound conjecture.

The main results of the present paper appear in Section 3. Theorem 3.1, which concerns the behavior of linear functionals on polytopes, may be of intrinsic interest, and it leads by means of polarity and Lemma 2.1 to the strong restrictions on shelling order contained in Theorem 3.2. These restrictions are currently being used, in the manner indicated at the end of Section 3, in a computational attack on several problems concerning the combinatorial structure of polytopes.

Section 1 contains two results on shellings of more general complexes.

1. Shellings of complexes. A complex $\mathcal{C}$ is here called a *combinatorial n -ball* provided that $\mathcal{C}$ admits a subdivision $\mathcal{S}$ combinatorially equivalent to a simplicial

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decomposition $\mathfrak{D}$ of an n -simplex; that is, there are isomorphic simplicial complexes $\mathfrak{S}$ and $\mathfrak{D}$ such that $\cup \mathfrak{S} = \cup \mathfrak{C}$, every cell of $\mathfrak{S}$ is contained in a cell of $\mathfrak{C}$, and $\cup \mathfrak{D}$ is an n -polytope with $n + 1$ vertices. *Combinatorial n -spheres* are similarly defined with respect to the boundary of an $(n + 1)$ -simplex. Note that $\mathfrak{C}$ is not required to be simplicial in these definitions.

The sort of semishelling defined in the Introduction is here called a *topological semishelling*, and two other sorts of semishelling are also considered. Let $(F_1, \dots, F_m)$ be an ordering of the facets of an n -complex $\mathfrak{C}$, for $1 \leq j \leq m$ let $\mathfrak{C}_j$ denote the complex consisting of $F_1, \dots, F_j$ and all faces of those n -cells, and for $1 < j \leq m$ let $\mathfrak{K}_j = \mathfrak{C}_{j-1} \cap \mathfrak{B}(F_j)$. Then $(F_1, \dots, F_m)$ is a *piecewise linear semishelling* of $\mathfrak{C}$ provided that for $1 < j \leq m$ either $\mathfrak{K}_j = \mathfrak{B}(F_j)$ or $\mathfrak{K}_j$ is a combinatorial $(n - 1)$ -ball. And $(F_1, \dots, F_m)$ is an *inductive semishelling* of $\mathfrak{C}$ provided that for $1 < j \leq m$ either $\mathfrak{K}_j = \mathfrak{B}(F_j)$ or $\cup \mathfrak{K}_j$ is a topological $(n - 1)$ -ball and $\mathfrak{K}_j$ is inductively semishellable. (To start the inductive definition it is agreed that for 1-complexes inductive and topological semishellings are identical.) *Piecewise linear shellings* and *inductive shellings* are defined in the obvious way.

The following two results are proved in tandem.

1.1 PROPOSITION. *For an arbitrary n -complex $\mathfrak{C}$, inductive semishellings are piecewise linear and piecewise linear semishellings are topological. The three sorts of semishelling coincide when $\mathfrak{C}$ is simplicial and also when $n \leq 3$.*

1.2 PROPOSITION. *If $(F_1, \dots, F_m)$ is a piecewise linear semishelling of an n -complex $\mathfrak{C}$ in which no $(n - 1)$ -cell is incident to more than two n -cells, then $(F_1, \dots, F_m)$ is a shelling and $\mathfrak{C}$ is a combinatorial n -ball or combinatorial n -sphere.*

Proofs. It is obvious that piecewise linear semishellings are topological. Now for each n let B_n denote the assertion of 1.2 and A_n the assertion that each inductive shelling of an n -complex is piecewise linear. Since A_1 and B_1 are obvious, we can establish A_n and B_n for all n by showing that $A_{n-1} \wedge B_{n-1} \Rightarrow A_n \wedge B_n$ when $n \geq 2$. Suppose, then, that A_{n-1} and B_{n-1} are known, and consider an inductive semishelling $(F_1, \dots, F_m)$ of an n -complex $\mathfrak{C}$. For $1 < j \leq m$ either $\mathfrak{K}_j = \mathfrak{B}(F_j)$ or $\cup \mathfrak{K}_j$ is a topological $(n - 1)$ -ball and $\mathfrak{K}_j$ admits an inductive semishelling $(G_1, \dots, G_r)$. It follows from A_{n-1} that $(G_1, \dots, G_r)$ is piecewise linear and then from B_{n-1} that $\mathfrak{K}_j$ is a combinatorial $(n - 1)$ -ball. But then $(F_1, \dots, F_m)$ is piecewise linear and $A_{n-1} \wedge B_{n-1} \Rightarrow A_n$.

We observe next that $A_{n-1} \wedge B_{n-1} \Rightarrow B_n^1$, where B_n^1 is the assertion that for each n -polytope P the complex $\mathfrak{C}(P) = \mathfrak{B}(P) \cup \{P\}$ is a combinatorial n -ball. Indeed, Bruggesser and Mani [7, p. 202] show $\mathfrak{B}(P)$ admits an inductive (hence by A_{n-1} a piecewise linear) shelling, whence $\mathfrak{B}(P)$ is a combinatorial $(n - 1)$ -sphere by B_{n-1} and then plainly $\mathfrak{C}(P)$ is a combinatorial n -ball.

To establish B_n let $(F_1, \dots, F_m)$ be as in 1.2 and let l denote the largest integer $r \leq m$ such that $\mathfrak{K}_j \neq \mathfrak{B}(F_j)$ for all $j \leq r$. Then $\mathfrak{C}_l$ is a combinatorial

n -ball, as follows from B_n^1 in conjunction with repeated application of the fact that if $\mathfrak{X}$ and $\mathfrak{Y}$ are combinatorial n -balls in the same complex and $\mathfrak{X} \cap \mathfrak{Y}$ is a combinatorial $(n - 1)$ -ball, then $\mathfrak{X} \cup \mathfrak{Y}$ is a combinatorial n -ball (Hudson [11, p. 39]). That completes the proof of B_n when $l = m$.

If $l < m$ then $\mathfrak{B}(F_{l+1})$ is a subcomplex of the boundary complex $\mathfrak{B}(\mathfrak{C}_l)$, and since $\cup \mathfrak{B}(F_{l+1})$ and $\cup \mathfrak{B}(\mathfrak{C}_l)$ are both topological $(n - 1)$ -spheres it is clear $\mathfrak{B}(F_{l+1}) = \mathfrak{B}(\mathfrak{C}_l)$ and consequently $l = m - 1$. Let $\mathfrak{S}$ be the boundary complex of an $(n + 1)$ -simplex and form $\mathfrak{J}$ from $\mathfrak{S}$ by omitting a single facet F . Since $\mathfrak{C}_l$ is a combinatorial n -ball and $\mathfrak{J}$ is combinatorially equivalent to a stellar subdivision of an n -simplex, there is a simplicial subdivision of $\mathfrak{C}_l$ that is equivalent to a simplicial subdivision of $\mathfrak{J}$. Under the latter equivalence a subdivision of $\mathfrak{B}(\mathfrak{C}_l) = \mathfrak{B}(F_m)$ corresponds to a subdivision of $\mathfrak{B}(\mathfrak{J}) = \mathfrak{B}(F)$, and hence by adding an additional vertex interior to F_m it is easy to obtain a subdivision of $\mathfrak{C}$ that is equivalent to a subdivision of $\mathfrak{S}$. That completes the proof of 1.2.

Note that if $\cup \mathfrak{K}$ is a topological k -ball then $\mathfrak{K}$ is inductively shellable for $k \leq 2$ (Newman [14], Frankl [8] and Sanderson [16, pp. 913–914]) and also when $\mathfrak{K}$ is a subcomplex of the boundary complex of a $(k + 1)$ -simplex. Hence all topological semishellings of an n -complex $\mathfrak{C}$ are inductive for $n \leq 3$ and also when $\mathfrak{C}$ is simplicial. That completes the proof of 1.1. $\square$

We do not know whether, in cases not covered by 1.1, topological shellings must be piecewise linear or piecewise linear shellings must be inductive. That a complex $\mathfrak{C}$ is shellable if $\cup \mathfrak{C}$ is a topological 2-sphere follows from the fact (Steinitz and Rademacher [17], Grünbaum [9], Barnette and Grünbaum [5]) that $\mathfrak{C}$ is then combinatorially equivalent to the boundary complex of a 3-polytope. Newman [14], Frankl [2], Rudin [15], and Grünbaum (private communication) have produced unshellable complexes $\mathfrak{C}$ such that $\cup \mathfrak{C}$ is a topological 3-ball; Rudin's example is actually a simplicial decomposition of a tetrahedron and Grünbaum's is especially simple in some other respects. We do not know whether all combinatorial 3-spheres are shellable, though it seems unlikely.

2. Shellings associated with linear functionals. The following is a basic tool for producing shellings of a polytope P that satisfy strong conditions on the order of appearance of P 's facets. It is merely a polar rephrasing of the main result of Bruggesser and Mani [7].

2.1 LEMMA. *Suppose P is a polytope in E^d with $0 \in \text{int } P$, and Q is the polar of P . Let V denote the set of all vertices of Q , and for each $v \in V$ let $F(v)$ be the corresponding facet of P . If $(v_1, \dots, v_m)$ is an ordering of V that is strictly monotone with respect to some linear functional ϕ , then $(F(v_1), \dots, F(v_m))$ is an inductive shelling of the boundary complex of P .*

Proof. By hypothesis,

$$P = \{x \in E^d : \langle x, q \rangle \leq 1 \text{ for all } q \in Q\},$$

and for each $v \in V$,

$$F(v) = \{p \in P : \langle p, v \rangle = 1\}.$$

We may assume without loss of generality that $0 \notin \phi V$ and ϕ is strictly decreasing on $(v_1, \dots, v_m)$. Let j be such that

$$\phi(v_1) > \dots > \phi(v_i) > 0 > \phi(v_{i+1}) > \dots > \phi(v_m)$$

and let $x \in E^d$ be such that $\phi(y) = \langle x, y \rangle$ for all $y \in E^d$. Since

$$\left\langle \frac{1}{\phi(v)} x, v \right\rangle = \frac{1}{\phi(v)} \langle x, v \rangle = 1,$$

the hyperplane containing $F(v)$ intersects the line Rx at the point $\mu(v)x$, where $\mu(v) = 1/\phi(v)$. But then $0 < \mu(v_1) < \mu(v_2) < \dots < \mu(v_i)$ and $\mu(v_{i+1}) < \mu(v_{i+2}) < \dots < \mu(v_m) < 0$, whence it follows from the reasoning of Bruggesser and Mani [7, pp. 202–203] that $(F(v_1), \dots, F(v_m))$ is an inductive shelling. $\square$

When an n -complex $\mathcal{C}$ is simplicial a given ordering $(F_1, \dots, F_m)$ of its facets can easily be tested to see whether it is a shelling or semishelling. For note that if $\mathcal{K}$ is a subcomplex of the boundary complex $\mathcal{B}(S)$ of an n -simplex S , then $\cup \mathcal{K}$ is a topological $(n-1)$ -ball if and only if $\mathcal{K}$ is an $(n-1)$ -complex different from $\mathcal{B}(S)$. It follows that $(F_1, \dots, F_m)$ is a semishelling if and only if the following conditions (a)–(b) hold for $1 \leq j \leq m$ and a shelling if and only if, in addition, (c) holds for $1 < j < m$, where X_i denotes the set of all vertices of F_i :

- (a) there exists $i < j$ such that $X_i \cap X_j \neq \emptyset$;
- (b) for each $i < j$ there exists $i' < j$ with $|X_{i'} \cap X_j| = n$ and $X_i \cap X_j \subset X_{i'} \cap X_j$;
- (c) there is an n -tuple in X_j that is not contained in X_i for any $i < j$.

If $\mathcal{C}$ is the boundary complex of an $(n+1)$ -polytope $P \subset E^{n+1}$ with $0 \in \text{int } P$, Q is the polar of P , and v_i is the vertex of Q corresponding under polarity to the facet F_i of P ($1 \leq i \leq m$), then conditions (a)–(c) translate under polarity into the following conditions:

- ($\bar{a}$) there exists $i < j$ such that v_i and v_j are on the same facet of Q ;
 - ($\bar{b}$) for each $i < j$ there exists $i' < j$ such that $[v_{i'}, v_j]$ is an edge of Q contained in the smallest face of Q that contains both v_i and v_j ;
 - ($\bar{c}$) v_i belongs to an edge of Q whose other end is not equal to v_i for any $i < j$.
- It is obvious without reference to [7] that ($\bar{a}$)–($\bar{c}$) are satisfied when the ordering $(v_1, \dots, v_m)$ arises from a linear functional, thus providing an alternative proof of 2.1 in the simplicial case.

3. A theorem on shelling order. We construct linear functionals subject to certain conditions and then deduce the corresponding shelling result with the aid of 2.1. The same method yields other results on shelling order but attention is confined here to the one that appears most useful. There is in principle no need to rely on the polar approach to shellings, but 2.1 is useful

because linear functionals are more familiar and easily manipulated than shellings *per se*.

3.1 THEOREM. *Suppose $v_0, v_1, \dots, v_d$ is a sequence in the vertex-set V of a d -polytope Q in E^d and*

$$\{v_0\} = G_0 \subset G_1 \subset \dots \subset G_d$$

is a sequence of faces of Q with $\dim G_i = i$ and $v_i \in G_i \sim G_{i-1}$. For $2 \leq i \leq d$ let

$$V_i = (V \cap G_i) \sim (G_{i-1} \cup \{v_i\})$$

and let A_i denote the members of V_i that are adjacent to v_i . Then E^d admits a linear functional ϕ and Q permits a nonsingular projective transformation T such that the restriction of ϕT to V is one-to-one and with respect to increasing ϕT the members of V are ordered as follows:

$$(*) \quad v_0, v_1, V_2, v_2, \dots, V_{d-1}, v_{d-1}, V_d, v_d.$$

Further, for each i such that v_i is i -valent in G_i the set A_i may be required to appear in $()$ at the end of V_i and in an arbitrary preassigned order.*

Proof. For $1 \leq i \leq d$ let γ_i denote the i^{th} coordinate functional on E^d —that is, $\gamma_i(x_1, \dots, x_d) = x_i$. Then we assert $A(d)$: Whenever $(Q, V, v_0, \dots, v_d, G_0, \dots, G_d)$ are as described there exists a nonsingular projective transformation T of Q such that for $1 \leq i \leq d$ the following conditions are satisfied:

- (a_i) $\gamma_i(TQ) = [0, 1]$, $(TQ) \cap \gamma_i^{-1}(0) = TG_{i-1}$, $(TQ) \cap \gamma_i^{-1}(1) = \{v_i\}$;
- (b_i) the restriction of $\gamma_i T$ to $V_i \cup \{v_i\}$ is one-to-one;
- (c_i) if v_i is i -valent in G_i then $\max \gamma_i T(V_i \sim A_i) < \min \gamma_i T A_i$ and the members of A_i are ordered by $\gamma_i T$ in an arbitrary preassigned manner.

The assertion $A(1)$ is obvious. Now suppose $A(d-1)$ is known and consider a system $(Q, V, v_0, \dots, v_d, G_0, \dots, G_d)$ satisfying the hypotheses of 3.1. With E denoting the subspace of E^d consisting of all points whose last coordinate is zero and π the transformation projecting E^d orthogonally onto E , we show in the following paragraphs that there exists a nonsingular projective transformation T_d of Q such that the following two conditions hold:

(i) the set $\pi T_d(Q \sim G_{d-1})$ is interior to the set $\pi T_d G_{d-1} = T_d G_{d-1}$ relative to the subspace E ;

(ii) conditions ((a_d), (b_d) and (c_d)) are satisfied when T is replaced by T_d . By the inductive hypothesis $A(d-1)$ there is a nonsingular projective transformation T_{d-1} of $T_d G_{d-1}$ in E such that

(iii) when T is replaced by T_{d-1} and Q by $T_d G_{d-1}$, conditions (a_i), (b_i) and (c_i) are all satisfied for $1 \leq i \leq d$. Now with $u = (0, \dots, 0, 1) \in E^d$, extend T_{d-1} to E^d by defining

$$T_{d-1}'(x) = T_{d-1}(x_1, \dots, x_{d-1}, 0) + x_d u$$

for all $x = (x_1, \dots, x_d) \in E^d$ and then set $T = T_{d-1}' \circ T_d$. By (iii), conditions

(a_d) , (b_d) and (c_d) are true as stated, and it follows from (i) in conjunction with (iii) that (a_i) , (b_i) , and (c_i) are true for $1 \leq i \leq d$. Thus to establish $A(d)$ for all d it remains only to construct the transformation T_d satisfying (i) and (ii).

Let H_0 denote the hyperplane containing G_{d-1} , and let $\mathfrak{F}$ denote the set of all $(d-2)$ -flats F in H_0 such that F misses G_{d-1} and F is contained in a hyperplane that strictly separates v_d from the remaining vertices of Q . For each $F \in \mathfrak{F}$, let $\mathcal{H}(F)$ denote the set of all hyperplanes that contain F and intersect Q . Let $H_0(F) = H_0$, let $H_1(F)$ be the member of $\mathcal{H}(F)$ that includes v_d , and for $0 < \lambda < 1$ let $H_\lambda(F)$ be the member of $\mathcal{H}(F)$ such that λ is the ratio of dihedral angle $(H_0(F), H_\lambda(F))$ to the dihedral angle $(H_0(F), H_1(F))$ (both angles being measured in $\mathcal{H}(F)$). Since Q is simply covered by the intersections $Q \cap H$ as H ranges over $\mathcal{H}(F)$, we may define a function ζ_F on Q by setting $\zeta_F(x) = \lambda$ when $x \in Q \cap H_\lambda(F)$. Note that the restriction of ζ_F to $V \sim G_{d-1}$ is one-to-one if and only if the following condition is satisfied:

(†) F neither intersects nor is parallel to any line determined by two members of $V \sim G_{d-1}$.

Note that the set $\mathfrak{F}$ is nonempty and is open relative to the space of all $(d-2)$ -flats contained in H_0 . Further, for each $F \in \mathfrak{F}$ there is a hyperplane that contains F and misses Q , whence Q permits a nonsingular projective transformation U_F that carries F into the hyperplane at infinity and therefore carries $H_0(F)$ and $H_1(F)$ into parallel hyperplanes. Under U_F the level sets of ζ_F in Q are carried into the level sets of a linear functional whose minimum and maximum sets in $U_F Q$ are $U_F G_{d-1}$ and $U_F(v_d)$ respectively.

The transformation T_d is obtained as a composition, $T_d = U_3 \circ U_2 \circ U_1$. In describing the construction, the images of a set X under U_1 , $U_2 \circ U_1$, and $U_3 \circ U_2 \circ U_1$ are denoted by X' , X'' , and X''' respectively. The next paragraph shows how to construct U_1 .

If the valence of v_d in Q exceeds d , and also if the set A_d is empty, choose an arbitrary member of $\mathfrak{F}$, perturb it slightly to obtain $F \in \mathfrak{F}$ for which condition (†) is satisfied, and then let $U_1 = U_F$. If the valence of v_d in Q is equal to d and the set A_d is nonempty let J be the hyperplane containing the d vertices of Q that are adjacent to v_d . It is not hard to verify that if $|A_d| = d$ then $J \cap H_0 = F_1 \in \mathfrak{F}$, while if $0 < |A_d| < d$ there is a $(d-2)$ -flat M such that $A_d \subset M \subset J$ and by rotating J slightly about M we obtain a hyperplane J_1 such that $J_1 \cap H_0 = F_1 \in \mathfrak{F}$. Since A_d is affinely independent it is then a routine matter to perturb F_1 slightly to obtain $F \in \mathfrak{F}$ such that condition (†) is satisfied and the set A_d is ordered in an arbitrary preassigned manner by the function ζ_F . Then set $U_1 = U_F$.

Now since G_{d-1}' is a facet of Q' , and G_{d-1}' and v_d' lie in parallel supporting hyperplanes of Q' , there exist $\epsilon > 0$ and a rigid motion U_2 of E^d such that $G_{d-1}'' \subset E$, the origin is interior to G_{d-1}'' relative to E , and $\gamma_d Q'' = [0, \eta]$ with $Q'' \cap \gamma_d^{-1}(\eta) = \{v_d\}$. (For the last use the fact that the maximum of ζ_F on Q is attained uniquely at v_d .)

For a sufficiently small $\xi > 0$ the point $-\xi u$ of E^d is beneath all facets of Q'' except for G_{d-1}'' and is of course beyond G_{d-1}'' . Hence all of Q'' except the relative boundary of G_{d-1}'' is interior to the convex cone K formed by the open rays that issue from $-\xi u$ and pass through the various points of G_{d-1}'' . Now for all $x \in E^d$ with $x_d > -\xi$ let

$$U_3(x) = \frac{\xi + \eta}{\eta(\xi + x_d)} x.$$

The nonsingular projective transformation U_3 is permissible for Q'' and carries the rays that form K onto a system of rays parallel to the ray $]0, \infty[u$. Hence all of the polytope Q''' except for G_{d-1}''' is interior to the cylinder $G_{d-1}''' +]0, \infty[u$, whence plainly condition (i) is satisfied. And since hyperplanes of the form " $\gamma_d = \text{constant} > 0$ " are carried by U_3 into other such hyperplanes, it is plain that (ii) holds. We have now shown the transformation T exists as described, and will complete the proof of 3.1 by constructing the linear functional ϕ .

For $2 \leq i \leq d$ choose $\delta_i > 0$ such that $\delta_i < \gamma_i(x)$ for all $x \in V_i \cup \{v_i\}$ and $\delta_i < |\gamma_i(x) - \gamma_i(y)|$ for all $x, y \in V_i \cup \{v_i\}$ with $x \neq y$. Let $\phi_1 = \gamma_1$, $\mu_1 = 1$, and (for $2 \leq i \leq d$) when ϕ_{i-1} and μ_{i-1} have been defined let

$$(*) \quad \phi_i = \phi_{i-1} + (2\mu_{i-1}/\delta_i)\gamma_i \quad \text{and} \quad \mu_i = \max \phi_i V.$$

Finally, let $\phi = \phi_d$. Since $G_0 \subset G_1 \subset \cdots \subset G_d$ and $\gamma_j G_i = \{0\}$ for $j > i$, it is clear that the ϕ -ordering of $V \cap G_i$ is the same as the ϕ_i -ordering. And it follows from $(*)$ and the definition of δ_i that the ϕ_i -ordering of $V_i \cup \{v_i\}$ is the same as the γ_i -ordering, while

$$\min \phi_i(V_i \cup \{v_i\}) > \max \phi_i(V \cap G_{i-1}).$$

But then the ϕ -ordering of V is as claimed and the proof of 3.1 is complete. $\square$

3.2 THEOREM. *Suppose P is a d -polytope,*

$$K_{d-1} \supset K_{d-2} \supset \cdots \supset K_0 \supset K_{-1} = \emptyset$$

is a sequence of faces of P with $\dim K_i = i$, and

$$K_{d-1} = F_0, F_1, \dots, F_d$$

is a sequence of facets of P such that F_i contains K_{d-1-i} but not K_{d-i} ($1 \leq i \leq d$). For $2 \leq i \leq d$ let $\mathfrak{F}_i$ denote the set of all facets J of P such that $J \supset K_{d-1-i}$, $J \supsetneq K_{d-i}$, and $J \neq F_i$. Then the boundary complex $\mathfrak{B}(P)$ admits an inductive shelling in which the facets of P appear in the following order:

$$F_0, F_1, \mathfrak{F}_2, F_2, \dots, \mathfrak{F}_{d-1}, F_{d-1}, \mathfrak{F}_d, F_d.$$

Further, if P is simplicial it may be required for $2 \leq i \leq d$ that the last members of $\mathfrak{F}_i$ in the ordering are those that have a $(d-2)$ -face in common with F_i , and they may be required to appear in an arbitrary preassigned order.

Proof. Assume without loss of generality that $0 \in \text{int } P$, and let Q denote the polar of P . Then apply 2.1 and 3.1, with G_i denoting the face of Q that corresponds to K_{d-1-i} under the polar correspondence, and v_i the vertex of Q that corresponds to F_i . $\square$

We close with an indication of how Theorem 3.2 can be used in computational attacks on combinatorial problems concerning polytopes. Suppose, for example, it is desired to generate the set $\mathfrak{B}_d$ of all combinatorial types of boundary complexes of simplicial d -polytopes with $2d$ vertices. That is easy when $d = 3$, but when $d \geq 4$ considerable ingenuity is required, partly because of the increased number of types and partly because of the lack of an intrinsic combinatorial characterization of the structures in question. (With the aid of a computer, Grünbaum and Sreedharan [10] solved the problem for $d = 4$. See also Brückner [6], Barnette [4], Altshuler [1], and Altshuler and Steinberg [2, 3].) A possible approach is to begin by generating, rather than $\mathfrak{B}_d$, a class $\mathfrak{S}_d$ of simplicial $(d - 1)$ -spheres with $2d$ vertices such that $\mathfrak{S}_d \supset \mathfrak{B}_d$ and membership in $\mathfrak{S}_d$ can be characterized in intrinsic combinatorial terms. The shellable spheres are a natural choice for $\mathfrak{S}_d$. However, difficulties may be expected because of the large number of combinatorial types, the large number of different shelling orders associated with a given type, and the fact that, even after the members of $\mathfrak{S}_d$ have been catalogued, it must be determined which of them can be realized as boundary complexes of polytopes. All of the difficulties can be mitigated somewhat by defining $\mathfrak{S}_d$ not in terms of mere shellability but in terms of the strongest conditions on shelling order that are known to be satisfied by boundary complexes of polytopes. At present those conditions are provided by Theorem 3.2.

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